



Storm Sequence Generator — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 26-March-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs (“MKM”).

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an “as is” basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

1 Test Results — Storm Sequence Generator (MKM-SS-001)

Tests run on **26 March 2026 at 18:58:34**.

Table 1: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

Total Tests	52
Passed	52
Failed	0
Skipped	0
Duration	0.17s

Table 2: Detailed Test Results — Storm Sequence Generator

Test Description	Acceptance Criteria	Result	Time
Duration must not exceed MAX_PRECIP_END_HOUR (156).	<code>result.storms[0].duration_hours <= MAX_PRECIP_END_HOUR</code>	Pass	0.000s
Isolated has one storm	<code>isinstance(result, StormSequence);</code> <code>result.num_storms == 1 (+2 more)</code>	Pass	0.000s
All storm sequences	<code>isinstance(seq, StormSequence)</code>	Pass	0.000s
Catchment propagated	<code>seq.catchment_id == "rhine";</code> <code>storm.catchment_id == "rhine"</code>	Pass	0.001s
Correct count	<code>len(small_batch) == 200</code>	Pass	0.000s
Different seeds differ	<code>types_a != types_b</code>	Pass	0.002s
Force sequence type doublet	<code>doublets > 50; seq.sequence_type in ("doublet", "isolated")</code>	Pass	0.002s
Force sequence type isolated	<code>seq.sequence_type == "isolated";</code> <code>seq.num_storms == 1</code>	Pass	0.001s
Reproducible with same seed	<code>[s.sequence_id for s in a] != [s.sequence_id for s in b];</code> <code>sa.num_storms == sb.num_storms</code> (+1 more)	Pass	0.003s
Returns list	<code>isinstance(small_batch, list)</code>	Pass	0.000s
Custom weight heavy on extreme shifts the distribution.	<code>all(c == "extreme" for c in cats)</code>	Pass	0.007s
All four default weight categories appear in the batch.	<code>cat in categories</code>	Pass	0.000s
Multi-storm sequences appear at a meaningful rate for severe+.	<code>multi_frac > 0.15</code>	Pass	0.000s
Creates parent dirs	<code>out.exists()</code>	Pass	0.008s
Load missing file raises	—	Pass	0.000s
Load returns correct count	<code>len(loaded) == len(small_batch)</code>	Pass	0.015s

Test Description	Acceptance Criteria	Result	Time
Load wrong schema raises	—	Pass	0.001s
Loaded sequences still valid	<code>result["invalid"] == 0</code>	Pass	0.016s
Round trip num storms	<code>orig.num_storms == back.num_storms</code>	Pass	0.016s
Round trip precipitation	<code>abs(orig.total_precipitation_mm - back.total_precipitatio...</code>	Pass	0.016s
Round trip sequence type	<code>orig.sequence_type == back.sequence_type</code>	Pass	0.015s
Round trip storm fields	<code>os_.storm_index == bs_.storm_index; abs(os_.duration_hours - bs_.duration_hours) < 0.01</code> (+1 more)	Pass	0.016s
Round trip total duration	<code>abs(orig.total_duration_hours - back.total_duration_hours...</code>	Pass	0.015s
Save creates file	<code>out.exists()</code>	Pass	0.007s
Schema version in file	<code>raw["schema_version"] == SCHEMA_VERSION</code>	Pass	0.015s
Creates file	<code>out.exists()</code>	Pass	0.000s
Duration stats present	<code>d["min"] > 0; d["max"] <= 156</code> (+1 more)	Pass	0.001s
Num sequences matches	<code>data["num_sequences"] == len(small.batch)</code>	Pass	0.001s
Precipitation stats present	<code>p["min"] > 0; p["max"] >= p["min"]</code> (+1 more)	Pass	0.001s
Schema version	<code>data["schema_version"] == SCHEMA_VERSION</code>	Pass	0.001s
Type counts sum to total	<code>sum(data["sequence_type_counts"].values()) == len(small.b...</code>	Pass	0.001s
Type fractions sum to one	<code>abs(total_frac - 1.0) < 0.001</code>	Pass	0.001s
Aggregate stats	<code>result.max_intensity_factor >= result.avg_intensity_factor; result.total_precipitation_mm > 0</code> (+1 more)	Pass	0.000s
Catchment id in storms	<code>result.storms[0].catchment_id == "rhine"</code>	Pass	0.000s
Cluster sequence	<code>result.sequence_type in (SequenceType.CLUSTER.value, Sequ...</code>	Pass	0.000s
force_sequence_type='doublet' → always doublet.	<code>result.sequence_type == SequenceType.DOUBLET.value; result.num_storms == 2</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
force_sequence_type='isolated' → always isolated.	<code>result.sequence_type ==</code> <code>SequenceType.ISOLATED.value;</code> <code>result.num_storms == 1</code>	Pass	0.000s
Minimal intensity (5% sequence probability) mostly produces isolated.	<code>isolated >= 10</code>	Pass	0.001s
Generate multiple sequences without error.	<code>isinstance(result, StormSequence);</code> <code>result.num_storms >= 1</code>	Pass	0.000s
Persistent sequence	<code>result.sequence_type in</code> <code>(SequenceType.PERSISTENT.value,</code> <code>S...</code>	Pass	0.000s
Returns storm sequence	<code>isinstance(result, StormSequence)</code>	Pass	0.000s
All sequences must have total_duration ≤ 168h.	<code>result.total_duration_hours <= 168.0</code>	Pass	0.000s
Storms have correct fields	<code>storm.duration_hours > 0;</code> <code>storm.intensity_factor > 0</code> (+2 more)	Pass	0.000s
Custom catchment	<code>gen.catchment_id == "rhine";</code> <code>gen.base_precipitation ==</code> <code>CATCHMENT_BASE_PRECIP["rhine"]</code>	Pass	0.000s
Default catchment	<code>gen.catchment_id == "thames";</code> <code>gen.base_precipitation ==</code> <code>CATCHMENT_BASE_PRECIP["thames"]</code>	Pass	0.000s
Same seed → same sequence.	<code>seq1.num_storms == seq2.num_storms;</code> <code>seq1.sequence_type ==</code> <code>seq2.sequence_type</code>	Pass	0.000s
Unknown catchment defaults to 35	<code>gen.base_precipitation == 35.0</code>	Pass	0.000s
Zero validation errors across the whole batch.	<code>result["invalid"] == 0</code>	Pass	0.000s
Medium batch all valid	<code>result["invalid"] == 0</code>	Pass	0.001s
No negative durations	<code>storm.duration_hours > 0</code>	Pass	0.000s
No negative precipitation	<code>seq.total_precipitation_mm > 0;</code> <code>storm.precipitation_mm > 0</code>	Pass	0.000s
Last storm precipitation must end by hour 156 (1e-9 FP tolerance).	<code>last_end <= MAX_PRECIP_END_HOUR +</code> <code>_FP_TOL</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
12-Mar-2026	1.0	Initial test suite (116 tests)	D. Kelly
26-Mar-2026	1.1	64 test(s) removed (total: 52)	D. Kelly